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(consequently $E^{-i} = E^i$, $G^{-i} = G^i$, $H^{-i} = H^i$, $L^{-i+k} = L^i$, $S^{-i+k} = S^i$, $T^{-i+k} = T^i$), and that the terms in question, putting in the coefficients $= \theta\theta' = 0$, are with him

$$\begin{aligned} & \{(1)^i + (11)^i\eta^2 + (17)^i\eta^4 + (20)^i\eta^6\} \cos(W - i\lambda), \\ & + \{(212)^i\eta^2 + (218)^i\eta^4 + (221)^i\eta^6\} \cos[W - (i-2)\lambda - 2\varpi'], \\ & + \{(372)^i\eta^4 + (375)^i\eta^6\} \cos[W - (i-4)\lambda - 4\varpi'], \\ & + \{(449)^i\eta^6\} \cos[W - (i-6)\lambda - 6\varpi'], \end{aligned}$$

where, substituting for $(1)^i$, $(11)^i$, &c., their values, the coefficients are

$$\begin{aligned} \frac{1}{4}A^i &= \eta^i \frac{1}{4}E^i + \eta^4 \cdot \frac{1}{4}G^i - \eta^6 \frac{1}{4}H^i; \\ -\frac{1}{4}A^i &= \eta^2 \cdot \frac{1}{4}(B^{i-1} + B^{i+1}) + \eta^4 \cdot \frac{1}{16}(C^{i-2} + 4C^i + C^{i+2}) - \eta^6 \cdot \frac{1}{64}(D^{i-3} + 9D^{i-1} + 9D^{i+1} + D^{i+3}); \\ \eta^2 \cdot \frac{1}{4}B^{i-1} - \eta^4 \cdot L^i + \eta^6 S^i, &= \eta^4 \cdot \frac{1}{8}L^{i-1} - \eta^6 \frac{1}{4}(D^{i-2} + C^i) + \eta^6 \cdot \frac{1}{16}(D^{i-3} + 3D^{i-1} + D^{i+1}); \\ \eta^4 \cdot \frac{1}{8}C^{i-2} - \eta^6 T^i, &= \eta^6 \cdot \frac{1}{8}C^{i-1} - \frac{1}{16}(D^{i-3} + D^{i-1}); \end{aligned}$$

and

$$\eta^6 \cdot \frac{1}{64}D^{i-3}.$$

Writing herein j in place of i , and for A^j , B^{j-1} , &c., the equal values A^{-j} , B^{-j+1} , &c., we have precisely the foregoing coefficients $D(j, -j), \dots D(j, -j+6)$.

The Development in Powers of e , e' .

The complete expression of the reciprocal of the distance is obtained from

$$\begin{aligned} & D(j, -j) \cos(jU - jU') \\ & + 2D(j, -j+2) \cos\{jU + (-j+2)U'\} \\ & + 2D(j, -j+4) \cos\{jU + (-j+4)U'\} \\ & + 2D(j, -j+6) \cos\{jU + (-j+6)U'\}, \end{aligned}$$

by writing therein for r , r' , U , U' , instead of the circular, the elliptic values, that is the values

$$\begin{aligned} r &= a \cdot \text{elqr}(e, L - \Pi), = a(1 + x), \\ r' &= a' \cdot \text{elqr}(e', L' - \Pi'), = a'(1 + x'), \\ U &= \Pi - \Theta + \text{elta}(e, L - \Pi), = U - \Theta + f, \\ U' &= \Pi' - \Theta' + \text{elta}(e', L' - \Pi'), = U' - \Theta' + f'; \end{aligned}$$

L , Π , Θ = the mean longitude in orbit, longitude of perihelion in orbit, and longitude of node; and the like for L' , Π' , Θ' ; “elqr”=elliptic quotient radius, “elta”=elliptic true anomaly; or, what is the same thing, if we write $\text{elta}(e, L - \Pi) = L - \Pi + \text{eltt}(e, L - \Pi)$, and the like for $\text{elta}(e', L' - \Pi')$, then

$$\begin{aligned} U &= L - \Theta + \text{eltt}(e, L - \Pi), = L - \Theta + y, \\ U' &= L' - \Theta' + \text{eltt}(e', L' - \Pi'), = L' - \Theta' + y'. \end{aligned}$$

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The process for doing this is explained, First Part, pp. 205–207, [214], viz., writing $r = a(1 + x)$, $r' = a'(1 + x')$, and restoring j' (instead of its value $-j, \dots -j+6$, as the case may be), we have a general term

$$D(j, j') x^\alpha x'^{\alpha'} \cos[jU + j'U'],$$

where $D(j, j')$ now denotes the value obtained by writing a, a' in place of r, r' and f, f' are the true anomalies $\text{elta}(e, L - \Pi)$ and $\text{elta}(e', L' - \Pi')$. And the second factor, $x^\alpha x'^{\alpha'}$ into the cosine, is given as a series

$$\Sigma \Sigma ([\cos]^\alpha + [\sin]^\alpha) ([\cos]^{\alpha'} + [\sin]^{\alpha'}) \cos[i(L - \Pi) + i'(L' - \Pi') + j(\Theta - \Theta') - j'(\Theta' - \Theta')],$$

where $[\cos]^\alpha, [\sin]^\alpha$ are functions of e , $[\cos]^{\alpha'}, [\sin]^{\alpha'}$ functions of e' . Or, what is better, the term $x^\alpha x'^{\alpha'}$ into the cosine may be written $x^\alpha x'^{\alpha'} \cos[j(L - \Theta + y) + j'(L' - \Theta' + y')]$, and the expansion then is

$$\Sigma \Sigma ([\cos]^\alpha + [\sin]^\alpha) ([\cos]^{\alpha'} + [\sin]^{\alpha'}) \cos[i(L - \Pi) + i'(L' - \Pi') + j(L - \Theta) + j'(L' - \Theta')],$$

where as before $[\cos]^\alpha, [\sin]^\alpha$ are functions of e , $[\cos]^{\alpha'}, [\sin]^{\alpha'}$ are the same functions of e' , viz. the e -functions are those given in the two “datum-tables” $(\alpha^0 \dots id') \cos jy$ and $(\alpha^0 \dots \alpha s') \sin jy$, taken from Leverrier, which I have given in my “Tables of the Developments of Functions in the Theory of Elliptic Motion,” *Memoirs R.A.S.* vol. xxix. (1861), pp. 191–306, [216]. In order to better show which are the symbols referred to, we may, instead of $[\cos]^\alpha$, &c., write $[x^\alpha \cos jy]^\alpha$, &c., the formula will then be

$$\begin{aligned} & x^\alpha x'^{\alpha'} \cos[j(L - \Theta + y) + j'(L' - \Theta' + y')] \\ &= \Sigma \Sigma ([x^\alpha \cos jy]^\alpha + [x^\alpha \sin jy]^\alpha) ([x'^{\alpha'} \cos j'y']^{\alpha'} + [x'^{\alpha'} \sin j'y']^{\alpha'}) \\ & \quad \times \cos[i(L - \Pi) + i'(L' - \Pi') + j(L - \Theta) + j'(L' - \Theta')], \end{aligned}$$

and if we attribute to i, i' any given values, that is, attend to any particular multiple cosine,

$$\cos[i(L - \Pi) + i'(L' - \Pi') + j(L - \Theta) + j'(L' - \Theta')],$$

the coefficient hereof will be

$$\Sigma \Sigma \frac{1}{1 \cdot 2 \dots \alpha} \alpha^\alpha \left(\frac{d}{d\alpha} \right)^\alpha \frac{1}{1 \cdot 2 \dots \alpha'} \alpha'^{\alpha'} \left(\frac{d}{d\alpha'} \right)^{\alpha'} D(j, j') \{ [x^\alpha \cos jy]^\alpha + [x^\alpha \sin jy]^\alpha \} \{ [x'^{\alpha'} \cos j'y']^{\alpha'} + [x'^{\alpha'} \sin j'y']^{\alpha'} \},$$

where α, α' each extend from zero to infinity, but to obtain the expression up to a given order p in e, e' , we take only the values up to $\alpha + \alpha' = p$.

Particular Case.

Thus, for instance, in $\cos[j(L - \Theta) - j'(L' - \Theta')]$ the terms independent of e' are

$$\begin{aligned} & D(j, -j) \{ [x^0 \cos jy]^0 + [x^0 \sin jy]^0 \} \\ & + \frac{1}{1} \alpha \frac{d}{d\alpha} D(j, -j) \{ [x \cos jy]^1 + [x \sin jy]^1 \} \\ & + \frac{1}{1 \cdot 2} \alpha^2 \left(\frac{d}{d\alpha} \right)^2 D(j, -j) \{ [x^2 \cos jy]^2 + [x^2 \sin jy]^2 \} \\ & + \&c. \end{aligned}$$

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which, observing that in the present case the sine terms vanish, is

1	$\frac{x^2}{8}$	$\frac{x^4}{384}$	$\frac{x^6}{46080}$	
1	$-8j^2$ $-54j^4$ $-3440j^6$	$+96j^2$ $+3920j^4$	$-1280j^2$	
	+4	$-48j^2 - 360j^4$	$-700j^2$	$\frac{1}{1} \cdot \alpha \frac{d}{d\alpha}$
	+4	$-96j^2 + 1920j^4$ $-1320j^6$		$\frac{1}{1 \cdot 2} \alpha^2 \left(\frac{d}{d\alpha} \right)^2$
		$+144 - 2880j^2$ $+144 - 5760j^2$		$\frac{1}{1 \cdot 2 \cdot 3} \alpha^3 \left(\frac{d}{d\alpha} \right)^3$ $\frac{1}{1 \cdot 2 \cdot 3 \cdot 4} \alpha^4 \left(\frac{d}{d\alpha} \right)^4$
			+14400 +14400	$\frac{1}{1 \cdot 2 \cdot \dots \cdot 5} \alpha^5 \left(\frac{d}{d\alpha} \right)^5$ $\frac{1}{1 \cdot 2 \cdot \dots \cdot 6} \alpha^6 \left(\frac{d}{d\alpha} \right)^6$ $\frac{1}{1 \cdot 2 \cdot \dots \cdot 7} \alpha^7 \left(\frac{d}{d\alpha} \right)^7$

viz. the term in e^0 is

$$e^0 \left\{ -j^2 + \frac{1}{2} \alpha \frac{d}{d\alpha} + \frac{1}{8} \alpha^2 \left(\frac{d}{d\alpha} \right)^2 \right\} D(j, -j),$$

viz. writing $\eta = 0$, and therefore $D(j, -j) = \frac{1}{2} A^{-j}$, the term in e^2 is

$$e^2 \left\{ -j^2 + \frac{1}{2} \alpha \frac{d}{d\alpha} + \frac{1}{8} \alpha^2 \left(\frac{d}{d\alpha} \right)^2 \right\} \frac{1}{2} A^{-j},$$

which, conformably with Leverrier's subscript notation

$$A_1^j = \frac{1}{1} \alpha \frac{d}{d\alpha} A^j, \quad A_2^j = \frac{1}{1 \cdot 2} \alpha^2 \left(\frac{d}{d\alpha} \right)^2 A^j, \text{ \&c.},$$

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I write

$$e^2 \left[-j^2 + \frac{1}{2} (1 + 2) A_1^{-j} + \frac{1}{8} \cdot 2 \cdot 2 A_2^{-j} \right] \frac{1}{2} A^{-j} + \frac{1}{2} A_1^{-j}.$$

The term in question is given by Leverrier as $(\frac{1}{2}e)^2(2)^i, = e^2 \cdot \frac{1}{4}(2), h = i$ and $K^i = A^i$,

$$= e^2 \cdot \frac{1}{4} \{ -2i^2 A^i + A_1^i + A_2^i \},$$

which agrees.

Similarly the term in e^4 is

$$\begin{aligned} & \frac{e^4}{384} [96j^2 - 54j^4 - 48j^2()_1 - 96j^4()_1 + 144()_2 + 144()_3] \frac{1}{2} A^{-j}, \\ & = \frac{e^4}{768} \{ (96j^4 - 54j^4) A^{-j} - 48j^2 A_1^{-j} - 96j^4 A_1^{-j} + 144 A_2^{-j} + 144 A_3^{-j} \}, \end{aligned}$$

and the term in question is given by Leverrier as $(\frac{1}{2}e)^4(4)^i, = e^4 \cdot \frac{1}{16}(4), h = i$ and $K^i = A^i$, which agrees. I have not made the comparison of any more terms.

Leverrier's Results expressed in terms of the Arguments, $L' - \Theta'$, $L' - \Pi'$, $L - \Theta$, $L - \Pi$.

The angles which Leverrier uses in his arguments are l' , λ , ϖ , ϖ' , and τ' , viz. we have,

$$\begin{aligned} l' &= \Theta' + (L' - \Theta'), \\ \lambda &= \Theta' + (L - \Theta), \\ \varpi &= \varpi' + (L' - \Theta') - (L' - \Pi'), \\ \varpi &= \Theta' + (L - \Theta) - (L - \Pi), \\ \tau' &= \Theta', \end{aligned}$$

where L , Π , Θ are the mean longitude of the planet m , its perihelion and the mutual node, all in the orbit of m ; and similarly L' , Π' , Θ' are the mean longitude of the planet m' , of its perihelion and of the mutual node, all in the orbit of m' . On substituting the foregoing values of l' , λ , &c., Θ' , as it should do, disappears, and the arguments are all of them linear functions of $L' - \Theta'$, $L' - \Pi'$, $L - \Theta$, $L - \Pi$; or, if we please, of $L' - \Theta'$, $L - L' + \Pi'$, $L - \Theta$, $L - \Pi$, that is of the distances of each planet from its own perihelion and from the mutual node. It is, I think, convenient to use these last angular distances, and accordingly in Leverrier's arguments, I write,

$$\begin{aligned} l' &= \Theta' + (L' - \Theta'), \\ \lambda &= \Theta' && + (L - \Theta), \\ \varpi &= \Theta' + (L - \Theta) - (L - \Pi), \\ \varpi' &= \Theta' + (L' - \Theta') - (L' - \Pi'), \\ \tau' &= \Theta', \end{aligned}$$

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and for the purpose of reference form as it were an Index to his result as follows:

Reciprocal of Distance = as follows

Terms of order zero: terms of orders 2, 4, 6, having the same arguments.

		$L' - \Theta'$	$L' - \Pi'$	$L - \Theta$	$L - \Pi$
$(1)^i$	(1 ... 20) cos	i	0	$-i$	0
$(21)^i(\frac{1}{2}e)(\frac{1}{2}e')$	(21 ... 30) ,,	i	+1	$-i$	-1
$(31)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^2$	(31 ... 34) ,,	i	+2	$-i$	-2
$(35)^i(\frac{1}{2}e)^3(\frac{1}{2}e')^3$	(35 ... 35) ,,	i	+3	$-i$	-3
$(36)^i(\frac{1}{2}e)^2\eta^2$	(36 ... 39) ,,	i	0	$-i + 2$	-2
$(40)^i(\frac{1}{2}e)(\frac{1}{2}e')\eta^2$	(40 ... 43) ,,	i	-1	$-i + 2$	-1
$(44)^i(\frac{1}{2}e')^2\eta^2$	(44 ... 47) ,,	i	-2	$-i + 2$	0
			+1	$-i + 2$	-3
$(48)^i(\frac{1}{2}e)^3(\frac{1}{2}e')\eta^2$	(48 ... 48) ,,	i			
$(49)^i(\frac{1}{2}e)(\frac{1}{2}e')^3\eta^2$	(49 ... 49) ,,	i	-3	$-i + 2$	+1

Terms of the first order: terms of orders 3, 5, 7, having the same arguments.

		$L' - \Theta'$	$L' - \Pi'$	$L - \Theta$	$L - \Pi$
$(50)^i \frac{1}{2}e$	50 (... 69) cos	i	0	$-i$	+1
$(70)^i \frac{1}{2}e'$	70 (... 89) ,,	i	+1	$-i$	0
$(90)^i (\frac{1}{2}e)^2 (\frac{1}{2}e')$	90 (... 99) ,,	i	+1	$-i$	-2
$(100)^i (\frac{1}{2}e)(\frac{1}{2}e')^2$	(100 ... 109) ,,	i	+2	$-i$	-1
$(110)^i (\frac{1}{2}e)^3 (\frac{1}{2}e')^2$	(110 ... 113) ,,	i	+2	$-i$	-3
$(114)^i (\frac{1}{2}e)^2 (\frac{1}{2}e')^3$	(114 ... 117) ,,	i	+3	$-i$	-2
$(118)^i (\frac{1}{2}e)^4 (\frac{1}{2}e')^3$	(118 ... 118) ,,	i	+3	$-i$	-4
$(119)^i (\frac{1}{2}e)^3 (\frac{1}{2}e')^4$	(119 ... 119) ,,	i	+4	$-i$	-3
$(120)^i (\frac{1}{2}e)\eta^2$	(120 ... 129) ,,	i	0	$-i + 2$	-1
$(130)^i (\frac{1}{2}e')\eta^2$	(130 ... 139) ,,	i	-1	$-i + 2$	0
$(140)^i (\frac{1}{2}e')^2 \eta^2$	(140 ... 143) ,,	i	0	$-i + 2$	-3
$(144)^i (\frac{1}{2}e)^2 (\frac{1}{2}e')\eta^2$	(144 ... 147) ,,	i	+1	$-i + 2$	-2
$(148)^i (\frac{1}{2}e)(\frac{1}{2}e')^2 \eta^2$	(148 ... 151) ,,	i	-1	$-i + 2$	-2
$(152)^i (\frac{1}{2}e)^2 \eta^2$	(152 ... 155) ,,	i	-2	$-i + 2$	-1
$(156)^i \frac{1}{2}e (\frac{1}{2}e')^2 \eta^2$	(156 ... 159) ,,	i	-2	$-i + 2$	+1
$(160)^i (\frac{1}{2}e')^3 \eta^2$	(160 ... 163) ,,	i	-3	$-i + 2$	0
$(164)^i (\frac{1}{2}e)^3 (\frac{1}{2}e')\eta^2$	(164 ... 164) ,,	i	+1	$-i + 2$	-4
$(165)^i (\frac{1}{2}e)^2 (\frac{1}{2}e')^2 \eta^2$	(165 ... 165) ,,	i	+2	$-i + 2$	-3
$(166)^i (\frac{1}{2}e)^2 (\frac{1}{2}e')^2 \eta^2$	(166 ... 166) ,,	i	-3	$-i + 2$	+2
$(167)^i (\frac{1}{2}e)(\frac{1}{2}e')^3 \eta^2$	(167 ... 167) ,,	i	-4	$-i + 2$	+1
$(168)^i (\frac{1}{2}e)\eta^4$	(168 ... 168) ,,	i	0	$-i + 2$	-3
$(169)^i (\frac{1}{2}e)^2 (\frac{1}{2}e')\eta^4$	(169 ... 169) ,,	i	-1	$-i + 4$	-2
$(170)^i (\frac{1}{2}e)(\frac{1}{2}e')^2 \eta^4$	(170 ... 170) ,,	i	-2	$-i + 4$	-1
$(171)^i (\frac{1}{2}e')^3 \eta^4$	(171 ... 171) ,,	i	-3	$-i + 4$	0

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Terms of second order: terms of orders 4, 6, having the same arguments.

		$L' - \Theta'$	$L' - \Pi'$	$L - \Theta$	$L - \Pi$
$(172)^i(\frac{1}{2}e)^2$	(172 ... 181) cos	i	0	$-i$	+2
$(182)^i(\frac{1}{2}e)(\frac{1}{2}e')$	(182 ... 191) ,,	i	+1	$-i$	+1
$(192)^i(\frac{1}{2}e')^2$	(192 ... 201) ,,	i	+2	$-i$	0
$(202)^i(\frac{1}{2}e)^3(\frac{1}{2}e')$	(202 ... 205) ,,	i	+1	$-i$	-3
$(206)^i(\frac{1}{2}e)(\frac{1}{2}e')^3$	(206 ... 209) ,,	i	+3	$-i$	-1
$(210)^i(\frac{1}{2}e)^4(\frac{1}{2}e')^2$	(210 ... 210) ,,	i	+2	$-i$	-4
$(211)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^4$	(211 ... 211) ,,	i	+4	$-i$	-2
$(212)^i\eta^2$	(212 ... 221) ,,	i	0	$-i + 2$	0
$(222)^i(\frac{1}{2}e)(\frac{1}{2}e')\eta^2$	(222 ... 225) ,,	i	+1	$-i + 2$	-1
$(226)^i(\frac{1}{2}e)(\frac{1}{2}e')\eta^2$	(226 ... 229) ,,	i	-1	$-i + 2$	+1
$(230)^i(\frac{1}{2}e)^2\eta^2$	(230 ... 230) ,,	i	0	$-i + 2$	-4
$(231)^i(\frac{1}{2}e)^2(\frac{1}{2}e')\eta^2$	(231 ... 231) ,,	i	-1	$-i + 2$	-3
$(232)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^2\eta^2$	(232 ... 232) ,,	i	-2	$-i + 2$	-2
$(233)^i(\frac{1}{2}e)(\frac{1}{2}e')^3\eta^2$	(233 ... 233) ,,	i	-3	$-i + 2$	-1
$(234)^i(\frac{1}{2}e')^4\eta^2$	(234 ... 234) ,,	i	-4	$-i + 2$	0
$(235)^i(\frac{1}{2}e')^3(\frac{1}{2}e)\eta^2$	(235 ... 235) ,,	i	+2	$-i + 2$	-2
$(236)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^2\eta^2$	(236 ... 236) ,,	i	-2	$-i + 2$	+2
$(237)^i(\frac{1}{2}e)^2\eta^4$	(237 ... 237) ,,	i	0	$-i + 4$	-2
$(238)^i(\frac{1}{2}e)(\frac{1}{2}e')\eta^4$	(238 ... 238) ,,	i	-1	$-i + 4$	-1
$(239)^i(\frac{1}{2}e')^2\eta^4$	(239 ... 239) ,,	i	-2	$-i + 4$	0

Terms of third order: terms of orders 5, 7, having the same arguments.

		$L' - \Theta'$	$L' - \Pi'$	$L - \Theta$	$L - \Pi$
$(240)^i(\frac{1}{2}e)^3$	(240 ... 249) cos	i	0	$-i$	+3
$(250)^i(\frac{1}{2}e)^2(\frac{1}{2}e')$	(250 ... 259) ,,	i	+1	$-i$	+2
$(260)^i(\frac{1}{2}e)(\frac{1}{2}e')^2$	(260 ... 269) ,,	i	+2	$-i$	+1
$(270)^i(\frac{1}{2}e')^3$	(270 ... 279) ,,	i	+3	$-i$	0

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Terms of third order (concluded):

		$L' - \Theta'$	$L' - \Pi'$	$L - \Theta$	$L - \Pi$
$(280)^i(\frac{1}{2}e)^4(\frac{1}{2}e')$	(280 ... 283) cos	i	+1	$-i$	-4
$(284)^i(\frac{1}{2}e)(\frac{1}{2}e')^4$	(284 ... 287) ,,	i	+4	$-i$	-1
$(288)^i(\frac{1}{2}e)^5(\frac{1}{2}e')^2$	(288 ... 289) ,,	i	+2	$-i$	-5
$(290)^i(\frac{1}{2}e)^3(\frac{1}{2}e')\eta^2$	(290 ... 299) ,,	i	0	$-i + 2$	+1
$(300)^i(\frac{1}{2}e')\eta^2$	(300 ... 309) ,,	i	+1	$-i + 2$	0
$(310)^i(\frac{1}{2}e)^2(\frac{1}{2}e')\eta^2$	(310 ... 313) ,,	i	-1	$-i + 2$	+2
$(314)^i(\frac{1}{2}e)(\frac{1}{2}e')^2\eta^2$	(314 ... 317) ,,	i	+2	$-i + 2$	-1
$(318)^i(\frac{1}{2}e)^2\eta^2$	(318 ... 318) ,,	i	0	$-i + 2$	-5
$(319)^i(\frac{1}{2}e)^2(\frac{1}{2}e')\eta^2$	(319 ... 319) ,,	i	-1	$-i + 2$	-4
$(320)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^2\eta^2$	(320 ... 320) ,,	i	-2	$-i + 2$	-3
$(321)^i(\frac{1}{2}e)(\frac{1}{2}e')^3\eta^2$	(321 ... 321) ,,	i	-3	$-i + 2$	-2
$(322)^i(\frac{1}{2}e)(\frac{1}{2}e')^4\eta^2$	(322 ... 322) ,,	i	-4	$-i + 2$	-1
$(323)^i(\frac{1}{2}e')^5\eta^2$	(323 ... 323) ,,	i	-5	$-i + 2$	0
$(324)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^3\eta^2$	(324 ... 324) ,,	i	-2	$-i + 2$	+3
$(325)^i(\frac{1}{2}e)^3(\frac{1}{2}e')^2\eta^2$	(325 ... 325) ,,	i	+3	$-i + 2$	-2
$(326)^i(\frac{1}{2}e)^3\eta^4$	(326 ... 329) ,,	i	0	$-i + 4$	-1
$(330)^i(\frac{1}{2}e')^3\eta^4$	(330 ... 333) ,,	i	-1	$-i + 4$	0
$(334)^i(\frac{1}{2}e)^2(\frac{1}{2}e')\eta^4$	(334 ... 334) ,,	i	+1	$-i + 4$	-2
$(335)^i(\frac{1}{2}e)(\frac{1}{2}e')^2\eta^4$	(335 ... 335) ,,	i	-2	$-i + 4$	+1

Terms of fourth order: terms of order 6, and of same argument.

		$L' - \Theta'$	$L' - \Pi'$	$L - \Theta$	$L - \Pi$
$(336)^i(\frac{1}{2}e)^4$	(336 ... 339) cos	i	0	$-i$	+4
$(340)^i(\frac{1}{2}e)^3(\frac{1}{2}e')$	(340 ... 343) ,,	i	+1	$-i$	+3
$(344)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^2$	(344 ... 347) ,,	i	+2	$-i$	+2
$(348)^i(\frac{1}{2}e)(\frac{1}{2}e')^3$	(348 ... 351) ,,	i	+3	$-i$	+1
$(352)^i(\frac{1}{2}e')^4$	(352 ... 355) ,,	i	+4	$-i$	0
$(356)^i(\frac{1}{2}e)^5(\frac{1}{2}e')$	(356 ... 356) ,,	i	+1	$-i$	-5
$(357)^i(\frac{1}{2}e)(\frac{1}{2}e')^5$	(357 ... 357) ,,	i	+5	$-i$	-1

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Terms of fourth order (concluded):

		$L' - \Theta'$	$L' - \Pi'$	$L - \Theta$	$L - \Pi$
$(358)^i (\frac{1}{2}e)^2 \eta^2$	(358 ... 358) cos	i	0	$-i + 2$	+2
$(362)^i (\frac{1}{2}e)(\frac{1}{2}e')\eta^2$	(362 ... 364) ,,	i	+1	$-i + 2$	+1
$(366)^i (\frac{1}{2}e')^2 \eta^2$	(366 ... 369) ,,	i	+2	$-i + 2$	0
$(370)^i (\frac{1}{2}e)^3 (\frac{1}{2}e')\eta^2$	(370 ... 370) ,,	i	-1	$-i + 2$	+3
$(371)^i (\frac{1}{2}e)(\frac{1}{2}e')^3 \eta^2$	(371 ... 371) ,,	i	+3	$-i + 2$	-1
$(372)^i \eta^4$	(372 ... 375) ,,	i	0	$-i + 4$	0
$(376)^i (\frac{1}{2}e)(\frac{1}{2}e')\eta^4$	(376 ... 376) ,,	i	+1	$-i + 4$	-1
$(377)^i (\frac{1}{2}e)(\frac{1}{2}e')\eta^4$	(377 ... 377) ,,	i	-1	$-i + 4$	+1

Terms of fifth order: terms of order 7 having the same arguments.

		$L' - \Theta'$	$L' - \Pi'$	$L - \Theta$	$L - \Pi$
$(378)^i (\frac{1}{2}e)^5$	(378 ... 381) cos	i	0	$-i$	+5
$(382)^i (\frac{1}{2}e)^4 (\frac{1}{2}e')$	(382 ... 385) ,,	i	+1	$-i$	+4
$(386)^i (\frac{1}{2}e)^3 (\frac{1}{2}e')^2$	(386 ... 389) ,,	i	+2	$-i$	+3
$(390)^i (\frac{1}{2}e)^2 (\frac{1}{2}e')^3$	(390 ... 393) ,,	i	+3	$-i$	+2
$(394)^i (\frac{1}{2}e)(\frac{1}{2}e')^4$	(394 ... 397) ,,	i	+4	$-i$	+1
$(398)^i (\frac{1}{2}e')^5$	(398 ... 401) ,,	i	+5	$-i$	0
$(402)^i (\frac{1}{2}e')^6 (\frac{1}{2}e)$	(402 ... 402) ,,	i	+1	$-i$	-6
$(403)^i (\frac{1}{2}e)(\frac{1}{2}e')^6$	(403 ... 403) ,,	i	+6	$-i$	-1
$(404)^i (\frac{1}{2}e)^3 \eta^2$	(404 ... 407) ,,	i	0	$-i + 2$	+3
$(408)^i (\frac{1}{2}e)^2 (\frac{1}{2}e')\eta^2$	(408 ... 411) ,,	i	+1	$-i + 2$	+2
$(412)^i (\frac{1}{2}e)(\frac{1}{2}e')^2 \eta^2$	(412 ... 415) ,,	i	+2	$-i + 2$	+1
$(416)^i (\frac{1}{2}e')^3 \eta^2$	(416 ... 419) ,,	i	+3	$-i + 2$	0
$(420)^i (\frac{1}{2}e)^4 (\frac{1}{2}e')\eta^2$	(420 ... 420) ,,	i	-1	$-i + 2$	+4
$(421)^i (\frac{1}{2}e)(\frac{1}{2}e')^4 \eta^2$	(421 ... 421) ,,	i	+4	$-i + 2$	-1
$(422)^i (\frac{1}{2}e)\eta^4$	(422 ... 425) ,,	i	0	$-i + 4$	+1
$(426)^i (\frac{1}{2}e')\eta^4$	(426 ... 429) ,,	i	+1	$-i + 4$	0
$(430)^i (\frac{1}{2}e)^2 (\frac{1}{2}e')\eta^4$	(430 ... 430) ,,	i	-1	$-i + 4$	+2
$(431)^i (\frac{1}{2}e)(\frac{1}{2}e')^2 \eta^4$	(431 ... 431) ,,	i	+2	$-i + 4$	-1
$(432)^i (\frac{1}{2}e)\eta^6$	(432 ... 432) ,,	i	0	$-i + 6$	-1
$(433)^i (\frac{1}{2}e')\eta^6$	(433 ... 433) ,,	i	-1	$-i + 6$	0

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Terms of sixth order.

		$L' - \Theta'$	$L' - \Pi'$	$L - \Theta$	$L - \Pi$
$(434)^i(\frac{1}{2}e)^6$	(434 ... 434) cos	i	0	$-i$	+6
$(435)^i(\frac{1}{2}e)^5(\frac{1}{2}e')$	(435 ... 435) ,,	i	+1	$-i$	+5
$(436)^i(\frac{1}{2}e)^4(\frac{1}{2}e')^2$	(436 ... 436) ,,	i	+2	$-i$	+4
$(437)^i(\frac{1}{2}e)^3(\frac{1}{2}e')^3$	(437 ... 437) ,,	i	+3	$-i$	+3
$(438)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^4$	(438 ... 438) ,,	i	+4	$-i$	+2
$(439)^i(\frac{1}{2}e)(\frac{1}{2}e')^5$	(439 ... 439) ,,	i	+5	$-i$	+1
$(440)^i(\frac{1}{2}e')^6$	(440 ... 440) ,,	i	+6	$-i$	0
$(441)^i(\frac{1}{2}e)^2\eta^2$	(441 ... 441) ,,	i	0	$-i + 2$	+4
$(442)^i(\frac{1}{2}e)^3(\frac{1}{2}e')\eta^2$	(442 ... 442) ,,	i	+1	$-i + 2$	+3
$(443)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^2\eta^2$	(443 ... 443) ,,	i	+2	$-i + 2$	+2
$(444)^i(\frac{1}{2}e)(\frac{1}{2}e')^3\eta^2$	(444 ... 444) ,,	i	+3	$-i + 2$	+1
$(445)^i(\frac{1}{2}e')^4\eta^2$	(445 ... 445) ,,	i	+4	$-i + 2$	0
$(446)^i(\frac{1}{2}e)^2\eta^4$	(446 ... 446) ,,	i	0	$-i + 4$	+2
$(447)^i(\frac{1}{2}e)(\frac{1}{2}e')\eta^4$	(447 ... 447) ,,	i	+1	$-i + 4$	+1
$(448)^i(\frac{1}{2}e')^2\eta^4$	(448 ... 448) ,,	i	+2	$-i + 4$	0
$(449)^i\eta^6$	(449 ... 449) ,,	i	0	$-i + 6$	0

Terms of seventh order.

		$L' - \Theta'$	$L' - \Pi'$	$L - \Theta$	$L - \Pi$
$(450)^i(\frac{1}{2}e)^7$	(450 ... 450) cos	i	0	$-i$	+7
$(451)^i(\frac{1}{2}e)^6(\frac{1}{2}e')$	(451 ... 451) ,,	i	1	$-i$	+6
$(452)^i(\frac{1}{2}e)^5(\frac{1}{2}e')^2$	(452 ... 452) ,,	i	2	$-i$	+5
$(453)^i(\frac{1}{2}e)^4(\frac{1}{2}e')^3$	(453 ... 453) ,,	i	3	$-i$	+4
$(454)^i(\frac{1}{2}e)^3(\frac{1}{2}e')^4$	(454 ... 454) ,,	i	4	$-i$	+3
$(455)^i(\frac{1}{2}e)^2(\frac{1}{2}e')^5$	(455 ... 455) ,,	i	5	$-i$	+2
$(456)^i(\frac{1}{2}e)(\frac{1}{2}e')^6$	(456 ... 456) ,,	i	6	$-i$	+1
$(457)^i(\frac{1}{2}e')^7$	(457 ... 457) ,,	i	7	$-i$	0
$(458)^i(\frac{1}{2}e)^3\eta^2$	(458 ... 458) ,,	i	0	$-i + 2$	+5

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Terms of seventh order (concluded).

		$L' - \Theta'$	$L' - \Pi'$	$L - \Theta$	$L - \Pi$
$(459)^i (\frac{1}{2}e)^3 (\frac{1}{2}e')\eta^2$	$(459 \dots 459) \cos$	i	$+1$	$-i + 2$	$+4$
$(460)^i (\frac{1}{2}e)^2 (\frac{1}{2}e')^2 \eta^2$	$(460 \dots 460) , ,$	i	$+2$	$-i + 2$	$+3$
$(461)^i (\frac{1}{2}e)^3 (\frac{1}{2}e')^3 \eta^2$	$(461 \dots 461) , ,$	i	$+3$	$-i + 2$	$+2$
$(462)^i (\frac{1}{2}e) (\frac{1}{2}e')^4 \eta^2$	$(462 \dots 462) , ,$	i	$+4$	$-i + 2$	$+1$
$(463)^i (\frac{1}{2}e')^5 \eta^2$	$(463 \dots 463) , ,$	i	$+5$	$-i + 2$	0
$(464)^i (\frac{1}{2}e)^3 \eta^4$	$(464 \dots 464) , ,$	i	0	$-i + 4$	$+3$
$(465)^i (\frac{1}{2}e)^2 (\frac{1}{2}e')\eta^4$	$(465 \dots 465) , ,$	i	$+1$	$-i + 4$	$+2$
$(466)^i (\frac{1}{2}e) (\frac{1}{2}e')^2 \eta^4$	$(466 \dots 466) , ,$	i	$+2$	$-i + 4$	$+1$
$(467)^i (\frac{1}{2}e')^3 \eta^4$	$(467 \dots 467) , ,$	i	$+3$	$-i + 4$	0
$(468)^i (\frac{1}{2}e)\eta^6$	$(468 \dots 468) , ,$	i	0	$-i + 6$	$+1$
$(469)^i (\frac{1}{2}e')\eta^6$	$(469 \dots 469) , ,$	i	$+1$	$-i + 6$	0

Here the several coefficients are ultimately given in terms of the before-mentioned quantities $A^i, B^i, C^i, D^i, E^i, G^i, H^i, L^i, S^i, T^i$ (functions of a, a'), and their differential coefficients in regard to a ,

$$\left(A_1^i = \frac{1}{1} \alpha \frac{d}{d\alpha} A^i, \quad A_2^i = \frac{1}{1.2} \alpha^2 \left(\frac{d}{d\alpha} \right)^2 A^i, \text{ \&c.} \right),$$

as follows:—we have Leverrier, pp. 299–330, a list of functions (1), (2), . . . (154) of the form

$$(1) = \frac{1}{2} K^i, \quad (2) = -2h^2 K^i + K_1^i + K_2^i, \quad (3) = -2v^2 K^i + K_1^i + K_2^i, \text{ \&c.},$$

involving i, h , and K^i and its derived functions K_1^i, K_2^i , &c. The coefficients of the several cosines are given by means of the functions in question, thus, first coefficient, above denoted as $(1)^i (1 \dots 20)$, is

$$(1)^i + (2)^i (\frac{1}{2}e)^2 + (3)^i (\frac{1}{2}e')^2 + \dots + (20)^i \eta^6$$

where $(1)^i = (1)$, $(2)^i = (2) \dots$ writing in the functions (1), (2) . . . (10), $h = i$, and $K^i = A^i$;

$$(11)^i = (1), \quad (12)^i = (2), \text{ \&c.}, \text{ writing } h = i \text{ and } K^i = E^i;$$

$$(20)^i = (1), \text{ writing } h = i \text{ and } K^i = -\frac{1}{4} S^i;$$

and so on for the various component coefficients $(1)^i, (2)^i, \dots (469)^i$.

But the resulting expressions, for the several integer values $i = -10$ to $+10$, are worked out in the Addition II. (*Numerical Tables for the Calculation of the Coefficients of the Development of the Disturbing Function*), pp. 358–383. And this Addition contains also, indicated by the letters δ and Δ respectively, the expressions of the

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terms which experience an alteration in passing from the development of the reciprocal of the distance to those of the disturbing functions m' upon m , and m upon m' respectively.

We have

Disturbing Function m' upon m

$$= m' \left\{ -\frac{r \cos H}{r'^2} + \frac{1}{\rho} \right\}.$$

Disturbing Function m upon m'

$$= m \left\{ -\frac{r' \cos H}{r^2} + \frac{1}{\rho} \right\}.$$

The expressions of $-\frac{r \cos H}{r'^2}$ and $-\frac{r' \cos H}{r^2}$, developed to the third order in the eccentricities and inclination, are given, Leverrier, pp. 272 and 274. Expressed in the terms of the foregoing arguments $L - \Theta'$, &c., and in terms of a, a' in place of a and $\mathbf{a}$, these are as follows:

$\frac{-r \cos H}{r'^2} = \frac{a}{a'^3}$ into		$L' - \Theta'$	$L' - \Pi'$	$L - \Theta$	$L - \Pi$
$-1 + \frac{3}{2}(e^2 + e'^2) + \eta^2$	cos	1	0	−1	0
$-e^2$	„	+1	+1	−1	−1
$+\frac{3}{4}e^2 - \frac{3}{4}ee'^2 - \frac{3}{4}e\eta^2$	„	+1	0	−1	+1
$-\frac{1}{4}e + \frac{3}{8}ee^2 + \frac{3}{8}e^3 + \frac{1}{4}e\eta^2$	„	+1	0	−1	−1
$-2e' + e^2e' + \frac{3}{8}e'^3 + 2e'\eta^2$	„				
$-\frac{3}{4}ee'^2$	„	+1	+1	−1	−2
$+\frac{1}{4}e'^2$	„	+1	+1	−1	−2
$-\frac{3}{4}e'^2$	„	−1	+2	+1	−1
$+\frac{1}{4}e'\eta^2$	„	+1	+2	−1	−1
	„	+1	0	+1	−1

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$\frac{-r \cos H}{r'^2} = \frac{a}{a'^3}$ into		$L' - \Theta'$	$L' - \Pi'$	$L - \Theta$	$L - \Pi$
$-\frac{1}{8}e^3$	cos	+1	0	−1	+2
$-\frac{3}{8}e^3$	”	+1	0	−1	−2
$+3ee'^2$	”			−1	+1
$-3e^2e'$	”		+1	+1	+1
	”		+2	−1	
$-\frac{15}{8}e^2e'$	”		−2	+1	−1
$-\frac{1}{4}e\eta^2$	”		+1	−1	+3
$-\frac{17}{8}e\eta^2$	”		+2	−1	
$-\frac{3}{4}e\eta^2$	”		+2	+1	−3
$+\frac{1}{8}e\eta^2$	”			−1	
	”		+3	+1	+2
$-\frac{1}{8}e^2e'$	”			−1	+1
	”		+3	+1	+1
$-\frac{1}{8}ee'^2$	”		+1		+1
$+\frac{1}{8}e^2e'$	”				+1
$-\frac{1}{8}e'^3$	”				
$-\frac{1}{8}e'\eta^2$	”	+1			
$-\frac{3}{8}e'\eta^2$	”				
$-2e'\eta^2$	”	+1		+	
	”				
	”		−		
	”		+		
	”		−		
	”		+		
	”		+1		
	”		+		

$\frac{r' \cos H}{r^2} = \frac{a'}{a^3}$ into		$L' - \Theta'$	$L' - \Pi'$	$L - \Theta$	$L - \Pi$
$-1 + \frac{3}{2}(e^2 + e'^2) + \eta^2$	cos	1	0	−1	0
$-e'^2$	”	+1	+1	−1	−1
$-2e + ee'^2 + \frac{3}{8}e^3 + 2e\eta^2$	”	+1	0	−1	−1
$+\frac{3}{4}e' - \frac{3}{4}e^2e' - \frac{3}{4}e\eta^2$					
$-\frac{1}{4}e' + \frac{3}{8}e^2e' + \frac{3}{8}e'^3 + \frac{1}{4}e'\eta^2$	”			+1	+1
$+\frac{1}{4}e\eta^2$	”		+	+1	−1
$-\frac{3}{4}e^2e'$	”	+2	−1	−2	+2
$-e^2$	”		+	+1	−1
	”	+1	+2	−1	−1
$+3e^2e'$	”				
	”	+1	−1	+1	
	”	+1	0	−1	+2
	”	+1		−1	−2
	”	−1	+1	+1	+1
	”		+2	+1	

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$\frac{r' \cos H}{r^2} = \frac{a'}{a^3}$ into		$L' - \Theta'$	$L' - \Pi'$	$L - \Theta$	$L - \Pi$
$-\frac{1}{8}e'^3$	cos	+1	0	+2	-1
$-\eta^2$	”		+		+1
$-\frac{1}{32}e^3$	”				
$-\frac{3}{8}e^3$	”			+	-1 + 3
	”				-1 - 3
	”		+		
$+\frac{15}{8}e^2e'$	”		-	+	+1 + 2
$-\frac{3}{4}e^2e'$	”	+		+	-1 + 2
$-\frac{1}{4}ee'^2$	”	-1	+2	+1	+1
$-\frac{1}{8}e'^3$	”		-	+3	+1
$-\frac{1}{8}e^3$	”				
	”	+1		+3	-1
$-2e\eta^2$	”		+		+1 + 1
$-\frac{1}{8}e'\eta^2$	”	+1		+1	+1

It is hardly necessary to observe that, to obtain the expressions of the Disturbing Functions, these additional terms are to be combined with the corresponding terms in the expression of the reciprocal of the distance: thus, in the Disturbing Function Ω (m' upon m), the entire term depending on $\cos[(L' - \Theta') - (L - \Theta)]$ is

$$= m' \cdot \{2(1, \dots 20)_{i=1} + \frac{a}{a'^3}(-1 + \frac{3}{2}(e^2 + e'^2) + \eta^2)\} \cos[(L' - \Theta') - (L - \Theta)],$$

where, however, the supplemental term is taken to the third order only.

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480.

ON THE EXPRESSION OF DELAUNAY'S l, g, h , IN TERMS OF HIS FINALLY ADOPTED CONSTANTS.

[From the *Monthly Notices of the Royal Astronomical Society*, vol. XXXII. (1871–72), pp. 8–16.]

We have in Delaunay's lunar theory,

l , the mean anomaly of the Moon,

g , the mean distance of perigee from ascending node,

h , the mean longitude of ascending node,

quantities which vary directly as the time, the coefficients of t , or values of $\frac{dl}{dt}, \frac{dg}{dt}, \frac{dh}{dt}$ being given in his *Théorie du Mouvement de la Lune*, vol. II. pp. 237, 238. But these values are not expressed in terms of his constants a (or n), e , γ , finally adopted as explained p. 800, and it seems very desirable to obtain the expressions of l, g, h , in terms of these finally adopted constants: I have accordingly effected this transformation (which I found less laborious than I had anticipated). It will be convenient to imagine the a, n, e, γ of pp. 237, 238 replaced by A, N, E, Γ respectively. This being so, and writing $\frac{n'}{n}$ for the $\frac{n'}{n}$ of p. 800 we have, p. 800,

$$A = a \left\{ 1 + \left[-\frac{2}{3}m^2 - \frac{223}{72}m^3 \right] \frac{n'^2}{n^2} + \left(-1 + \frac{3}{2}\gamma^2 + \frac{3}{2}e^2 - \frac{3}{2}e'^2 - \frac{15}{4}\gamma^4 + \frac{15}{4}\gamma^2e^2 - \frac{15}{4}\gamma^2e'^2 + \frac{15}{4}e^4 - \frac{15}{4}e^2e'^2 - \frac{15}{4}e'^4 \right) m^2 + \dots \right\}$$

[Long power-series expansion in γ, e, e', m for A , following pp. 237–238 and 800 of Delaunay; full coefficient string preserved in the displayed equation above; for full numerical detail consult the IA scan.]

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and hence calculating from the formula $N^2 A^3 = n^2 a^3$, we find

$$N = n\{1 + [\frac{1}{2}m^2 + \frac{61}{48}m^3]\frac{n'^2}{n^2} + \dots\},$$

$$\begin{aligned} N &= n + (-\frac{1}{2} + \frac{3}{4}\gamma^2 + \frac{3}{4}e^2 - \frac{3}{4}e'^2 - \frac{15}{8}\gamma^4 + \frac{15}{8}\gamma^2 e^2 - \frac{15}{8}\gamma^2 e'^2 + \frac{15}{8}e^4 - \frac{15}{8}e^2 e'^2 - \frac{15}{8}e'^4)m^2 + \dots \\ &= n(1 + Q) \text{ suppose.} \end{aligned}$$

The values of E , Γ are given p. 800, but for the present purpose we only require E^2 and Γ^2 to the fifth order, viz. the values of these are at once found to be

$$E^2 = e^2(1 + \dots), \quad \Gamma^2 = \gamma^2(1 + \dots),$$

whence also $E^4 = e^4$ and $\Gamma^4 = \gamma^4$.

The formulae of pp. 237–238 now give

$$\begin{aligned} \frac{dl}{dt} &= n + (\dots)(1 + Q)^{-1} \\ &\quad + (\dots)m^2(1 + Q)^{-2} \\ &\quad + (\dots)m^3(1 + Q)^{-3} \\ &\quad + (\dots)m^4(1 + Q)^{-4} \\ &\quad + (\dots)m^5(1 + Q)^{-5}. \end{aligned}$$

(Observe that writing herein $Q = 0$, and omitting the terms in m^4 and m^5 in the coefficient of $(1 + Q)^{-1}$, and the term in m^5 in the coefficient of $(1 + Q)^{-2}$, we have the original formula of p. 237.)

$$\frac{dg}{dt} = nt\{[\frac{1}{4}m^2 + \frac{19}{16}m^3]\frac{n'^2}{n^2} + \dots\}(1 + Q)^{-1}$$

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[The expansion continues with the analogous expressions for $\frac{dg}{dt}$ and $\frac{dh}{dt}$, expanded in powers of m in the form $(1+Q)^{-1}, (1+Q)^{-2}, \dots, (1+Q)^{-5}$ together with the “observe” parenthetical remarks that, on writing $Q = 0$ and omitting the highest-order coefficients of m , one recovers the original formulæ of Delaunay (pp. 237 and 238).]

We hence have

$$l = nt\{A + (1+B)Q + CQ^2\}, \text{ i.e. } l = nt\{A - \frac{1}{3}Q + BQ + CQ^2\},$$

$$g = nt\{A' + B'Q + C'Q^2\},$$

$$h = nt\{A'' + B''Q + C''Q^2\},$$

where (omitting the terms in $\frac{n'^2}{n^2}$)

$$\begin{aligned} A = & 1 + (-\frac{1}{4} + \frac{3}{4}\gamma^2 - \frac{1}{4}e^2 - \frac{1}{4}e'^2 - \frac{15}{16}\gamma^4 + \frac{15}{8}\gamma^2e^2 - \frac{15}{8}\gamma^2e'^2 + \frac{15}{8}e^4 - \frac{15}{8}e^2e'^2 - \frac{15}{8}e'^4)m^2 \\ & + (-\frac{215}{16} + \frac{75}{8}\gamma^2 - \frac{15}{8}e^2 - \frac{15}{8}e'^2 - \frac{105}{16}\gamma^4 + \frac{35}{8}\gamma^2e^2 + \frac{35}{8}\gamma^2e'^2 + \frac{75}{8}e^4 - \frac{35}{8}e^2e'^2 - \frac{75}{8}e'^4)m^3 + \dots \end{aligned}$$

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$$B = (\dots)m^2 + (\dots)m^3 + \dots$$

$$C = -\frac{1}{4}m^2 - \frac{105}{16}m^3 - \dots$$

$$\begin{aligned} A' = & (\frac{3}{4} - \frac{3}{4}\gamma^2 + \frac{1}{2}e^2 + \frac{1}{2}e'^2 - \frac{15}{8}\gamma^4 + \frac{15}{8}\gamma^2e^2 + \frac{15}{8}\gamma^2e'^2 - \frac{15}{8}\gamma^2e'^2 + \frac{15}{8}e^4 + \frac{15}{8}e^2e'^2 - \frac{15}{8}e'^4)m^2 \\ & + (\dots)m^3 + (\dots)m^4 \end{aligned}$$

$$B' = (-\frac{15}{16} + \frac{15}{8}\gamma^2 - \frac{3}{8}e^2 + \frac{3}{8}e'^2)m^2 + \dots$$

$$C' = \frac{3}{16}m^3 + \dots$$

$$A'' = (-\frac{1}{4} + \frac{3}{4}\gamma^2 - \frac{1}{4}e^2 - \frac{1}{4}e'^2 - \frac{15}{16}\gamma^4 + \frac{15}{8}\gamma^2e^2 + \frac{15}{8}\gamma^2e'^2 + \frac{15}{8}e^4 - \frac{15}{8}e^2e'^2 - \frac{15}{8}e'^4)m^2 + \dots$$

$$B'' = (\frac{1}{8} - \frac{3}{8}\gamma^2 + \frac{1}{8}e^2 + \frac{3}{8}e'^2)m^2 + \dots$$

$$C'' = -\frac{3}{16}m^3 + \dots$$

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And in terms $BQ, B'Q, B''Q$, we have

$$Q = (1 - \frac{1}{2}\gamma^2 + \frac{3}{4}e^2 + \frac{3}{4}e'^2)m^2 + \dots$$

and in the terms $CQ^2, C'Q^2, C''Q^2$, simply $= Q^2 = m^4$. Hence finally the required values of l, g, h , are

$$l = nt\{1 + [-\frac{1}{4}m^2 - \frac{215}{48}m^3]\frac{n'^2}{n^2}\}$$

$$+ (-\frac{1}{4} + \frac{3}{4}\gamma^2 + \frac{1}{2}e^2 - \frac{1}{4}e'^2 - \frac{15}{16}\gamma^4 - \frac{15}{8}\gamma^2e^2 + \frac{15}{8}\gamma^2e'^2 + \frac{15}{8}e^4 - \frac{15}{8}e^2e'^2 - \frac{15}{8}e'^4)m^2 + \dots$$

$$g = nt\{[\frac{3}{4}m^2 + \frac{61}{16}m^3]\frac{n'^2}{n^2} + \dots\}$$

$$+ (\frac{3}{4} - \frac{3}{4}\gamma^2 + \frac{1}{2}e^2 + \frac{1}{2}e'^2 + \frac{15}{8}\gamma^4 - \frac{15}{8}\gamma^2e^2 - \frac{15}{8}\gamma^2e'^2 + \frac{15}{8}e^4 - \frac{15}{8}e^2e'^2 + \frac{15}{8}e'^4)m^2 + \dots$$

$$h = nt\{[-\frac{3}{4}m^2 - \frac{61}{16}m^3]\frac{n'^2}{n^2}\}$$

$$+ (-\frac{1}{4} + \frac{3}{4}\gamma^2 - \frac{3}{4}e^2 - \frac{3}{4}e'^2 - \frac{15}{16}\gamma^4 + \frac{15}{8}\gamma^2e^2 + \frac{15}{8}\gamma^2e'^2 - \frac{15}{8}e^4 - \frac{15}{8}e^2e'^2 - \frac{15}{8}e'^4)m^2 + \dots$$

[p. 533]

which values satisfy, as they should do, the equation $l + g + h = nt$. I recall that the precise signification of the constants is as follows: n is the coefficient of t in the expression of the Moon's longitude in terms of the time, a the corresponding elliptic value of the mean distance ($n^2 a^3 = \text{sum of masses}$), e the eccentricity, such that in the expression of the longitude the coefficient of the leading term of the equation of the centre has its elliptic value

$$= 2e - \frac{1}{4}e^3 + \frac{5}{96}e^5,$$

and γ the sine of the half-inclination, such that in the expression of the latitude the coefficient of the leading term has its elliptic value

$$= 2\gamma - 2\gamma e^2 - \frac{1}{2}\gamma^4 + \frac{1}{6}\gamma^2 e^2 + \frac{1}{2}\gamma^3 e^2 - \frac{1}{6}\gamma e^4.$$

n' , a' are the mean motion and mean distance of the Sun, $m = \frac{n'}{n}$, and e' is the eccentricity of the Sun's orbit, considered as constant.

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481.

ON THE EXPRESSION OF M. DELAUNAY'S $h + g$ IN TERMS OF HIS FINALLY ADOPTED CONSTANTS.

[From the *Monthly Notices of the Royal Astronomical Society*, vol. XXXII. (1871–72), p. 74.]

I HAD the pleasure of receiving from M. Delaunay a letter dated Paris, 17th Dec. 1871, in which he informs me that, on referring to his papers, he had found there expressions for l , g , h , identical with those given by me in the November Number of the *Monthly Notices*, with only a single typographical error, $-\frac{15}{8}e'^2m^3$ instead of $-\frac{15}{8}e'^4m^3$ [*ante* p. 532, corrected] in my expression of A .

M. Delaunay mentions also that he had obtained four additional terms in the expression for $h + g$ (longitude of the Moon's perigee), and that the complete expression in terms of the finally adopted constants is

$$\begin{aligned} h + g = nt & \left\{ \left(\frac{1}{2} - Gy^2 - \frac{1}{2}e^2 + \frac{1}{4}e'^2 - \frac{15}{16}\gamma^4 + \frac{15}{8}\gamma^2e^2 + \frac{15}{8}\gamma^2e'^2 - 9\gamma^2e^4 - \frac{15}{8}e^4 - \frac{15}{8}e'^4 + \frac{15}{8}e^2e'^2 \right) m^2 \right. \\ & + (\dots)m^3 + (\dots)m^4 + (\dots)m^5 \\ & \left. + \left[\frac{1}{2}m^2 + \frac{61}{16}m^3 \right] \frac{n'^2}{n^2} \right\}. \end{aligned}$$

[Observe that $h + g$ is $= nt - l$, and compare with the expression for l , *ante* p. 532.]

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482.

NOTE ON A PAIR OF DIFFERENTIAL EQUATIONS IN THE LUNAR THEORY.

[From the *Monthly Notices of the Royal Astronomical Society*, vol. XXXII. (1871–72), pp. 31–32.]

THE equations

$$\frac{d}{dt} \frac{d\rho}{dt} - \rho \left(\frac{dv}{dt} \right)^2 + \frac{1}{\rho^2} = km^2 \rho \left\{ \frac{1}{2} + \frac{3}{2} \cos(2v - 2mt) \right\},$$

$$\frac{d}{dt} \rho^2 \frac{dv}{dt} = jm^2 \rho^2 \left\{ -\frac{3}{2} \sin(2v - 2mt) \right\},$$

taking therein $j = k = 1$ in effect present themselves in the Lunar Theory, and particular integrals in series have been obtained, the development being carried to a great extent; but I give the results only as far as m^4 , viz., writing

$$t - mt = D,$$

we have

$$v = t + \left(\frac{11}{8}m^2 + \frac{59}{12}m^3 \right) \sin 2D + \frac{281}{96}m^4 \sin 4D,$$

$$\frac{1}{\rho} = 1 + \frac{1}{6}m^2 - \frac{1}{12}m^4 + \left(\frac{2}{3}m^2 + \frac{5}{6}m^3 + \frac{17}{32}m^4 \right) \cos 2D + \frac{1}{3}m^4 \cos 4D.$$

In the Lunar Theory j and k are properly each $= \frac{1}{1 + \frac{E}{m'}}$ (E the mass of the Earth, m' that of the Sun),

but they are taken to be $= 1$; the numerical difference is inappreciable; but there would be a considerable theoretical advantage in retaining

[p. 536]

in the equations the coefficients j, k [regarded as each of them $= k$]: in fact, the developments could then be arranged according to the powers of k , that is according to the powers of the disturbing force; whereas, when k is taken $= 1$, we have only a development in powers of m , and since m also presents itself through the coefficient $-2m$ of t in $2v - 2mt$, terms which are really of different orders in regard to the disturbing force, are united together into a single term: so that, instead of a term of the form $(Ak + Bk^2 + \&c.)m^p$, where A, B , are numerical, we have the term $(A + B + \dots)m^p$, where of course $A + B + \dots$ is given as a single numerical coefficient.

There is no equal advantage in retaining the two coefficients k, j , as this only serves to show how a term arises from the central and tangential forces respectively; thus retaining these coefficients, the integrals as far as m^2 are

$$v = t + \left(\frac{3}{8}k + j \right) m^2 \sin 2D,$$

$$\frac{1}{\rho} = 1 + \frac{1}{6}m^2k + \left(\frac{1}{6}k + \frac{1}{2}j \right) m^2 \cos 2D,$$

agreeing with the former result when $k = j = 1$; but there is, nevertheless, some interest in retaining the two coefficients. I hope to develop the results somewhat further, and to communicate them to the Society.

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483.

ON A PAIR OF DIFFERENTIAL EQUATIONS IN THE LUNAR THEORY.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXXII. (1871–72), pp. 201–206.]

I CONSIDER the differential equations

$$\frac{d}{dt} \frac{d\rho}{dt} - \rho \left(\frac{dv}{dt} \right)^2 + \frac{1}{\rho^2} = km^2 \rho \left\{ \frac{1}{2} + \frac{3}{2} \cos(2v - 2mt) \right\},$$

$$\frac{d}{dt} \left(\rho^2 \frac{dv}{dt} \right) = jm^2 \rho^2 \left\{ -\frac{3}{2} \sin(2v - 2mt) \right\},$$

which when $j = k = 1$ give the following equations in the lunar theory ($D = t - mt$):

$$\begin{aligned} \frac{1}{\rho} = & 1 + \frac{1}{6}m^2 - \frac{1}{12}m^4 - \frac{11}{63}m^5 - \frac{1721}{432}m^6 - \frac{74129}{3456}m^7 - \frac{145}{1296}m^8 \\ & + \cos 2D \left[m^2 + \frac{5}{6}m^3 + \frac{17}{32}m^4 + \frac{89}{72}m^5 + (\dots)m^6 + (\dots)m^7 + (\dots)m^8 \right] \\ & + \cos 4D \left[\frac{1}{3}m^4 + \frac{43}{60}m^5 + (\dots)m^6 + (\dots)m^7 + (\dots)m^8 \right] \\ & + \cos 6D \left[(\dots)m^6 + (\dots)m^7 + (\dots)m^8 \right] \end{aligned}$$

or as far as m^5 ,

$$\begin{aligned} \frac{1}{\rho} = & 1 + \frac{1}{6}m^2 - \frac{1}{12}m^4 - \frac{11}{63}m^5 + \cos 2D \left[m^2 + \frac{5}{6}m^3 + \frac{17}{32}m^4 + \frac{89}{72}m^5 \right] \\ & + \cos 4D \left[\frac{1}{3}m^4 + \frac{43}{60}m^5 \right] \\ & + \cos 6D \left[(\dots)m^6 \right]. \end{aligned}$$

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($\frac{1}{\rho}$ is given by M. Delaunay only as far as m^5 , the additional terms of $\frac{1}{\rho}$ and expression for ρ were kindly communicated to me by Prof. Adams); and

$$\begin{aligned} & v = t \\ & + \sin 2D \left[\frac{11}{8}m^2 + \frac{59}{12}m^3 + \frac{1471}{48}m^4 + (\dots)m^5 + (\dots)m^6 + (\dots)m^7 + (\dots)m^8 \right] \\ & + \sin 4D \left[\frac{281}{96}m^4 + \frac{1199}{144}m^5 + (\dots)m^6 + (\dots)m^7 + (\dots)m^8 \right] \\ & + \sin 6D \left[(\dots)m^6 + (\dots)m^7 + (\dots)m^8 \right] \end{aligned}$$

(Delaunay, t. II. pp. 815, 836, 845).

To integrate the original equations write

$$\rho = 1 + \rho_1 + \rho_2 + \rho_3 + \dots,$$

$$v = t + v_1 + v_2 + v_3 + \dots,$$

where the suffixes indicate the degrees in the coefficients k, j conjointly: the equations for ρ_n, v_n take the form

$$\begin{aligned} \frac{d}{dt} \frac{d\rho_n}{dt} - 3\rho_n - 2\frac{dv_n}{dt} + V_n &= Q_n, \\ \frac{d}{dt} \left(\frac{dv_n}{dt} + 2\rho_n + U_n \right) &= P_n, \end{aligned}$$

where V_n, U_n, P_n, Q_n do not contain ρ_n or v_n . From the second equation we have

$$\frac{dv_n}{dt} + 2\rho_n + U_n = \Omega_n + \int P_n dt,$$

where Ω_n is a constant of integration, the integral $\int P_n dt$ containing only periodic terms; and then adding twice this to the first equation we have

$$\frac{d}{dt} \frac{d\rho_n}{dt} + \rho_n + V_n + 2U_n = 2\Omega_n + Q_n + 2 \int P_n dt,$$

which determines ρ_n ; and substituting its value in the other equation we have $\frac{dv_n}{dt}$ and thence v_n ; the constant Ω_n is determined so that $\frac{dv_n}{dt}$ may contain no constant term. We have

$$V_2 = - \left(\frac{dv_1}{dt} \right)^2 - 2\rho_1 \frac{dv_1}{dt} + 3\rho_1^2, \quad V_3 = 2\rho_1 \frac{dv_1}{dt} + (2\rho_2 + \rho_1^2) \frac{dv_1}{dt} - 2\rho_1 \rho_2,$$

$$U_2 = \rho_1 \frac{dv_1}{dt} - \rho_2 \frac{dv_1}{dt} - \rho_1 \left(\frac{dv_1}{dt} \right)^2, \quad \&c.$$

$$Q_2 = -2\rho_1 \frac{d\rho_1}{dt} + 6\rho_1 \rho_2 - 4\rho_1^3, \quad \&c.$$

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$$\begin{aligned} Q_1 &= km^2(\tfrac{1}{2} + \tfrac{3}{2}\cos 2D), \quad P_1 = jm^2(-\tfrac{3}{2}\sin 2D), \\ Q_2 &= km^2\{3\rho_1 \sin 2D + \rho_1(\tfrac{1}{2} + \tfrac{3}{2}\cos 2D)\}, \\ Q_3 &= km^2\{-3v_1 \sin 2D - 3v_1^2 \cos 2D + \rho_2(\tfrac{1}{2} + \tfrac{3}{2}\cos 2D)\}, \quad P_2 = jm^2(-3v_1 \cos 2D - 3\rho_1 \sin 2D), \\ P_3 &= jm^2\{-3v_2 \cos 2D + 3v_1^2 \sin 2D - 6\rho_1 v_1 \cos 2D + (2\rho_2 + \rho_1^2)(-\tfrac{3}{2}\sin 2D)\}, \\ &\quad \&c. \end{aligned}$$

In particular attending to the values of P_1, Q_1 the equations for ρ_1, v_1 are in their original form

$$\begin{aligned} \frac{d}{dt} \frac{d\rho_1}{dt} - 3\rho_1 + 2\frac{dv_1}{dt} &= km^2(\tfrac{1}{2} + \tfrac{3}{2}\cos 2D), \\ \frac{d}{dt} \left(\frac{dv_1}{dt} + 2\rho_1 \right) &= jm^2(-\tfrac{3}{2}\sin 2D), \end{aligned}$$

whence in the transformed form they are

$$\frac{dv_1}{dt} + 2\rho_1 = \Omega_1 + \tfrac{3}{4} \frac{jm^2}{1-m} \cos 2D,$$

and

$$\frac{d}{dt} \frac{d\rho_1}{dt} + \rho_1 = 2\Omega_1 + \tfrac{1}{2}km^2 + (\tfrac{3}{2}km^2 + \tfrac{3}{2} \frac{jm^2}{1-m}) \cos 2D.$$

Thus the constant term of ρ_1 is $2\Omega_1 + \tfrac{1}{2}km^2$, giving in $\frac{dv_1}{dt}$ a constant term $-3\Omega_1 - km^2$; this must vanish, or we have $\Omega_1 = -\tfrac{1}{3}km^2$; and the equations thus become

$$\begin{aligned} \frac{d}{dt} \frac{d\rho_1}{dt} + \rho_1 &= -\tfrac{1}{6}km^2 + \left(\tfrac{3}{2}km^2 + \tfrac{3}{2} \frac{jm^2}{1-m} \right) \cos 2D, \\ \frac{dv_1}{dt} + 2\rho_1 &= -\tfrac{1}{3}km^2 + \left(\tfrac{3}{4} \frac{jm^2}{1-m} \right) \cos 2D, \end{aligned}$$

and then completing the integration

$$\begin{aligned} \rho_1 &= -\tfrac{1}{6}km^2 + \left\{ \frac{-\tfrac{3}{2}km^2}{(3-8m+4m^2)} + \frac{-\tfrac{3}{4}jm^2}{(1-m)(3-8m+4m^2)} \right\} \cos 2D, \\ v_1 &= \left\{ \frac{\tfrac{3}{8}km^2}{(1-m)(3-8m+4m^2)} + \frac{\tfrac{3}{8}jm^2(7-8m+4m^2)}{(1-m)^2(3-8m+4m^2)} \right\} \sin 2D, \end{aligned}$$

which are the accurate values of ρ_1 and v_1 .

Expanding as far as m^5 we have

$$\begin{aligned} \rho_1 &= k(-\tfrac{1}{6}m^2) + \cos 2D \left[k(-\tfrac{1}{2}m^2 - \tfrac{4}{3}m^3 - \tfrac{37}{8}m^4 - \tfrac{43}{3}m^5) + j(-\tfrac{1}{4}m^2 - \tfrac{5}{6}m^3 - \tfrac{61}{32}m^4 - \tfrac{49}{6}m^5) \right] \\ &= k(-\tfrac{1}{6}m^2 - \tfrac{3}{4}m^3 - \tfrac{19}{32}m^4 - \tfrac{55}{12}m^5 - \tfrac{135}{16}m^5), \text{ which for } j = k \text{ is} \end{aligned}$$

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and

$$\begin{aligned} v_1 &= \sin 2D \left\{ k(\tfrac{1}{8}m^2 + \tfrac{4}{6}m^3 + \tfrac{17}{16}m^4 + \tfrac{73}{16}m^5) + j(\tfrac{3}{8}m^2 + \tfrac{2}{3}m^3 + \tfrac{47}{32}m^4 + \tfrac{31}{6}m^5 + \tfrac{47}{16}m^5) \right\} \\ &= k(\tfrac{11}{8}m^2 + \tfrac{29}{12}m^3 + \tfrac{47}{32}m^4 + \tfrac{49}{16}m^5 + \tfrac{47}{16}m^5), \text{ which for } j = k \text{ is} \end{aligned}$$

I have, not in general, but for the value $j = k$, calculated ρ_2 and v_2 as far as m^5 . I have not made the calculation for ρ_3 and v_3 , but their values may be deduced from the foregoing values of ρ , v ; the final expressions (when $j = k$) of $\rho = 1 + \rho_1 + \rho_2 + \rho_3 + \dots$ and $v = t + v_1 + v_2 + v_3 + \dots$ are

$$\begin{aligned} \rho &= 1 \\ &+ k(-\tfrac{1}{6}m^2) \\ &+ k^2(-\tfrac{1}{12}m^4) \\ &+ \cos 2D \left\{ k(-\tfrac{1}{2}m^2 - \tfrac{4}{3}m^3 - \tfrac{37}{8}m^4 - \tfrac{43}{3}m^5 - \tfrac{135}{16}m^5) + k^2(\tfrac{1}{32}m^4 + \tfrac{4}{3}m^5 + \tfrac{61}{16}m^5) + j(-\tfrac{5}{12}m^5) \right\} \\ &+ \cos 4D \left\{ k^2(\tfrac{1}{3}m^4 - \tfrac{4}{3}m^4 - \tfrac{8}{3}m^5 - \tfrac{37}{12}m^6) + k^3(\tfrac{8}{3}m^5 + \tfrac{61}{16}m^5) \right\} \\ &+ \cos 6D \{ k^3(-\tfrac{1}{6}m^6) \}, \end{aligned}$$

and

$$\begin{aligned} v &= t - \tfrac{1}{4}m^4 + \sin 2D \left\{ k(\tfrac{11}{8}m^2 + \tfrac{29}{12}m^3 + \tfrac{47}{32}m^4 + \tfrac{49}{16}m^5 + \tfrac{47}{16}m^5) + k^2(\tfrac{1}{8}m^4 - \tfrac{4}{3}m^4 - \tfrac{43}{8}m^5 - \tfrac{27}{16}m^5) \right\} \\ &+ \sin 4D \left\{ k^2(\tfrac{281}{96}m^4 + \tfrac{1199}{144}m^5 + \tfrac{47}{32}m^6) \right\} + \sin 6D \left\{ k^3(\tfrac{47}{96}m^6 + \tfrac{47}{32}m^7) \right\}, \end{aligned}$$

which for $k = 1$ agree with the foregoing formulae (verifying them as far as m^4); the present formulae exhibit the manner in which the expressions depend on the several powers of the disturbing force.

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484.

ON THE VARIATIONS OF THE POSITION OF THE ORBIT IN THE PLANETARY THEORY.

[From the *Monthly Notices of the Royal Astronomical Society*, vol. XXXII. (1871–72), pp. 206–211.]

IT has always appeared to me that in the Planetary Theory, more especially when the method of the variation of the elements is made use of, there is a difficulty as to the proper mode of dealing with the inclinations and longitudes of the nodes, hindering the ulterior development of the theory. Considering the case of two planets m , m' , and referring their orbits to any fixed plane and fixed origin of longitudes therein, let θ , θ' be the longitudes of the nodes, ϕ , ϕ' the inclinations ($p = \tan \phi \sin \theta$, $q = \tan \phi \cos \theta$, &c., as usual); then the disturbing functions for m , m' respectively are developed, not explicitly in terms of ϕ , ϕ' , θ , θ' , but in terms of Φ , the mutual inclination of the two orbits, and of Θ , Θ' the longitudes in the two orbits respectively of the mutual node of the two orbits; Φ and Θ , Θ' being functions (and complicated ones) of ϕ , ϕ' , θ , θ' . Moreover, although in the general theory of the secular variations of the orbits of the planetary system, θ , ϕ , &c., are, as above, referred to one fixed plane (the ecliptic of a certain date), yet in the theory of each particular planet it is the practice, and obviously the convenient one, to refer for such planet the θ , ϕ to its own fixed plane (the orbit of the planet at a certain date), the effect of course being that ϕ , and consequently p , q , instead of being of the order of the inclinations to the ecliptic, are only of the order of the disturbing forces. It has occurred to me that the last-mentioned plan should be adhered to throughout; viz., that for each planet m , the position of its variable orbit should be determined by θ , the longitude of its node, and ϕ , the inclination in reference to the appropriate fixed plane (orbit of the planet at a certain date) and origin of longitude therein. The disturbing functions for the planets m and m' will of course depend not only on θ , θ' , ϕ , ϕ' , but on the quantities Φ , Θ , Θ' which determine the mutual positions of the two fixed

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planes of reference and origins of longitude therein, these last being however absolute constants *not affected by any variation of the elements*; so that as regards the variation of the elements the disturbing functions are in fact given as explicit functions of the variable elements $\theta, \phi, \theta', \phi'$; and where ϕ, ϕ' and therefore also p, q, p', q' are only of the order of the disturbing forces.

I proceed to work out this idea, for the present considering the development of the Disturbing Function only as far as the first powers of p, q , &c. For comparison with the ordinary theory, observe that in this theory the disturbing function contains only the second powers of the p, q , &c., made use of therein; these are in fact of a form such as $P + p, Q + q, \dots$ where P, Q are absolute constants and $p, q, \dots$ are the $p, q, \dots$ of the present theory; the ordinary theory gives therefore in the disturbing function a series of terms involving $(P + p)^2, (P + p)(Q + q), \dots$ which I now take account of only as far as the first powers of $p, q, \dots$ viz., they are in effect reduced to $P^2 + 2Pp, PQ + Pq + Qp$, &c. $\dots$ The present theory is thus not now developed to the extent of giving the $p, q, \dots$ of the ordinary theory in the more complete form as the solutions of a system of simultaneous linear differential equations, but only to the extent of obtaining for these $p, q, \dots$ respectively the terms which are proportional to the time.

I commence with the following subsidiary problem. Consider a spherical triangle ABC (sides a, b, c , angles A, B, C , as usual), and taking the side c as constant, but the angles A and B as variable, let it be required to find the variations of C, a, b in terms of variations dA, dB and the variable elements C, a, b themselves. Although the geometrical proof would be more simple, I give the analytical one, as it may be useful.

We have

$$\cos C = -\cos A \cos B + \sin A \sin B \cos c,$$

and thence

$$\begin{aligned} -\sin C dC &= (\sin A \cos B + \cos A \sin B \cos c) dA + (\sin A \cos B + \sin A \cos B \cos c) dB \\ &= dA \sin b \frac{\sin c}{\tan b} + \sin A \frac{\sin c}{\tan a} dB, \end{aligned}$$

that is

$$\frac{\sin C dC}{\sin c} = -dA \frac{\sin b \cos B}{\sin b} + \frac{\sin A \cos a}{\sin a} dB,$$

or finally

$$-dC = \cos b dA + \cos a dB.$$

Next

$$\sin a = \frac{\sin c}{\sin C} \sin A,$$

or, differentiating,

$$\cos a da = \frac{\sin c}{\sin^2 C} (\sin C \cos A dA - \cos C \sin A dC),$$

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or, substituting for dC its value,

$$\begin{aligned} &= \frac{\sin c}{\sin^2 C} \{dA (\sin C \cos A + \cos C \sin A \cos b) + dB \cos C \sin A \cos a\}, \\ &= \frac{\sin c}{\sin^2 C} \left\{ dA \frac{\sin C \sin b \cos a}{\sin a} + dB \cos C \sin A \cos a \right\}, \end{aligned}$$

that is

$$\frac{\sin c}{\sin C} da = \left\{ dA \frac{\sin b}{\sin a} + dB \frac{\cos C \sin A}{\sin C} \right\},$$

or, on the right-hand writing $\frac{\sin A}{\sin a}$ instead of $\frac{\sin C}{\sin c}$, this is

$$da = \frac{1}{\sin C} (dA \sin b + dB \cos C \sin a);$$

and similarly

$$db = \frac{1}{\sin C} (dB \sin a + dA \cos C \sin b).$$

Now let the continuous lines represent the orbits of m, m' at certain dates, O, Q the origins of longitude therein; and the dotted lines the variable orbits of the planets respectively.

[Figure: spherical-triangle diagram showing the fixed orbits of m and m' meeting at C (mutual node), with arcs $OC = \Theta$, $QC = \Theta'$, $\angle C = \Phi$; the variable orbits of the two planets (dotted) meeting in C' , with $CA = \theta$, $CB = \theta'$, the inclinations being $\angle CAC' = \phi$, $\angle CBC' = \phi'$.]

Write

$$\begin{aligned} OQ = \Theta, \quad CA = \theta, \quad \angle CAC' = \phi, \\ QC = \Theta', \quad CB = \theta', \quad \angle CBC' = \phi', \\ \angle C = \Phi. \end{aligned}$$

Then, answering to the notation of the lemma, we have

$$a = \theta', \quad b = \theta, \quad C = \Phi, \quad dA = d\phi, \quad dB = -d\phi',$$

or say $\tan \phi = -\tan \phi'$, whence

$$\begin{aligned} C'B &= a + da, \\ &= \theta' + \frac{1}{\sin \Phi} (\tan \phi \sin \theta - \tan \phi' \cos \Phi \sin \theta'), \\ &= \theta' + \frac{1}{\sin \Phi} (p - p' \cos \Phi), \end{aligned}$$

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$$\begin{aligned}
 C'A &= b + db, \\
 &= \theta + \frac{1}{\sin \Phi} (-\tan \phi' \sin \theta' + \tan \phi \cos \Phi \sin \theta), \\
 &= \theta - \frac{1}{\sin \Phi} (p' - p \cos \Phi), \\
 \angle C' &= \Phi - q + q' = \Phi + \tan \phi \cos \theta - \tan \phi' \cos \theta', = \Phi - q + q'.
 \end{aligned}$$

Suppose v, v' are the longitudes of the planets in their two orbits respectively; that is

$$\begin{aligned}
 v &= OA + Am = \Theta + \theta + Am, \\
 v' &= QB + Bm' = \Theta' + \theta' + Bm',
 \end{aligned}$$

whence

$$\begin{aligned}
 C'm &= C'A + Am, = v - \Theta - \frac{1}{\sin \Phi} (p' - p \cos \Phi), \\
 C'm' &= C'B + Bm', = v' - \Theta' + \frac{1}{\sin \Phi} (p - p' \cos \Phi), \\
 \angle C &= \Phi - q + q'.
 \end{aligned}$$

Say these values are $v - \Theta + x, v' - \Theta' + x', \Phi + y$. Then if H is the angular distance mm' of the two planets,

$$\begin{aligned}
 \cos H &= \cos(v - \Theta + x) \cos(v' - \Theta' + x') + \sin(v - \Theta + x) \sin(v' - \Theta' + x') \cos(\Phi + y), \\
 &= \cos(v - \Theta) \cos(v' - \Theta') + \sin(v - \Theta) \sin(v' - \Theta') \cos \Phi \\
 &\quad + x [-\sin(v - \Theta) \cos(v' - \Theta') + \cos(v - \Theta) \sin(v' - \Theta') \cos \Phi] \\
 &\quad + x' [-\cos(v - \Theta) \sin(v' - \Theta') + \sin(v - \Theta) \cos(v' - \Theta') \cos \Phi] \\
 &\quad + y [-\sin(v - \Theta) \sin(v' - \Theta') \sin \Phi], \\
 &= \cos H + \nabla, \text{ suppose.}
 \end{aligned}$$

The disturbing function for the planet m disturbed by m' is

$$R = m' \left\{ \frac{1}{\sqrt{r^2 + r'^2 - 2rr' \cos H}} - \frac{r \cos H}{r'^2} \right\}$$

(R , if $\Omega = -R$, is the disturbing function of the *Mécanique Céleste*); and the term hereof which involves ∇ is

$$= \nabla \frac{d \cos H}{d \cos H} \left\{ \frac{rr'}{(r^2 + r'^2 - 2rr' \cos H)^{\frac{3}{2}}} + \frac{r}{r'^2} \right\} \nabla,$$

where after the differentiation $\cos H$ is replaced by $\cos H$,

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viz., this is a linear function of x, x', y , that is of p, q, p', q' , with coefficients which of course involve the other variable elements and the time; but it will be remembered that Θ, Θ', Φ are not variable elements, but are absolute constants. The variations of p depend upon $\frac{d\Omega}{dq}$ and those of q on $\frac{d\Omega}{dp}$, and the quantities $p, q, p', q', \dots$ disappear from these differential coefficients $\frac{d\Omega}{dq}, \frac{d\Omega}{dp}$; that is, disregarding periodic terms, and the variations of the elements, we obtain $\frac{dp}{dt}, \frac{dq}{dt}$ as absolute constants, or reckoning the time from the epoch belonging to the fixed orbit of m , we have p, q as mere multiples of the time ($p = At, q = Bt$, where A and B are constants); agreeing with the statement preceding the investigation.

Observe that the p, q , as used above, have reference not only to the fixed orbit of m , but also to the node thereon of the fixed orbit of m' : we may, if we please, write $p = \tan \phi \sin(\theta + \Theta), q = \tan \phi \cos(\theta + \Theta)$,

that is, $p = \mathfrak{p} \cos \Theta + \mathfrak{q} \sin \Theta$, $q = -\mathfrak{p} \sin \Theta + \mathfrak{q} \cos \Theta$ (or $\mathfrak{p} = p \cos \Theta - q \sin \Theta$, $\mathfrak{q} = p \sin \Theta + q \cos \Theta$), and in place of p, q introduce into the formulae $\mathfrak{p}$ and $\mathfrak{q}$, which have reference only to the fixed orbit of m , and similarly writing $p' = \tan \phi' \sin(\theta' + \Theta')$, $q' = \tan \phi' \cos(\theta' + \Theta')$, instead of p', q' introduce $\mathfrak{p}', \mathfrak{q}'$ which have reference only to the fixed orbit of m' .

I remark that a table for the relative positions of the orbits of the eight Planets for the Epoch 1st January, 1850, is given in Leverrier's *Annales de l'Observ. de Paris*, t. II. (1856), pp. 64–66.

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PROBLEMS AND SOLUTIONS.

[From the *Mathematical Questions with their Solutions from the Educational Times*, vols. v. to xii. (1866–1869).]

[Vol. v., January to July, 1866, p. 17.]

1791. (*Proposed by Professor Cayley.*) — Given a quartic curve $U = 0$, to find three cubic curves $P = 0$, $Q = 0$, $R = 0$, each meeting the quartic in the same six points 1, 2, 3, 4, 5, 6, and such that $P = 0$, $R = 0$ may besides meet the quartic in the same three points a, b, c , and that $Q = 0$, $R = 0$ may besides meet the quartic in the same three points α, β, γ .

[Vol. v., pp. 25, 26.]

Note on the Problems in regard to a Conic defined by five Conditions of Intersection.

I USE the word “intersection” rather than “contact” because it extends to the case of a 1-pointic intersection, which cannot be termed a contact. The conditions referred to are that the conic shall have with a given curve, at a point given or not given, a 1-pointic intersection, a 2-pointic intersection (= ordinary contact), a 3-pointic intersection, &c., as the case may be. It may be noticed that when the point on the curve is a given point, the condition of a k -pointic intersection is really only the condition that the conic shall pass through k given points; though from the circumstance that these are consecutive points on a conic, the formulae for a conic passing through k discrete points require material alteration; for instance, in the two questions to find the equation of a conic passing through five given points, and to find the equation of a conic having at a given point of a given curve 5-pointic intersection with the curve, the forms of the solutions are very different from each other.

The foregoing remark shows, however, that it is proper to detach the conditions which relate to intersections at given points; and consequently attending only to the